

مباركة طوبى 6
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Cache memory

- If the active portions of the program and data are placed in a fast small memory, the average memory access time can be reduced,
- Thus reducing the total execution time of the program
- Such a fast small memory is referred to as cache memory
- The cache is the fastest component in the memory hierarchy and approaches the speed of CPU component

Cache memory

- When CPU needs to access memory, the cache is examined
- If the word is found in the cache, it is read from the fast memory
- If the word addressed by the CPU is not found in the cache, the main memory is accessed to read the word

Cache memory

- The performance of cache memory is frequently measured in terms of a quantity called **hit ratio**
- When the CPU refers to memory and finds the word in cache, it is said to produce a **hit**
- Otherwise, it is a **miss**
- **Hit ratio = hit / (hit+miss)**

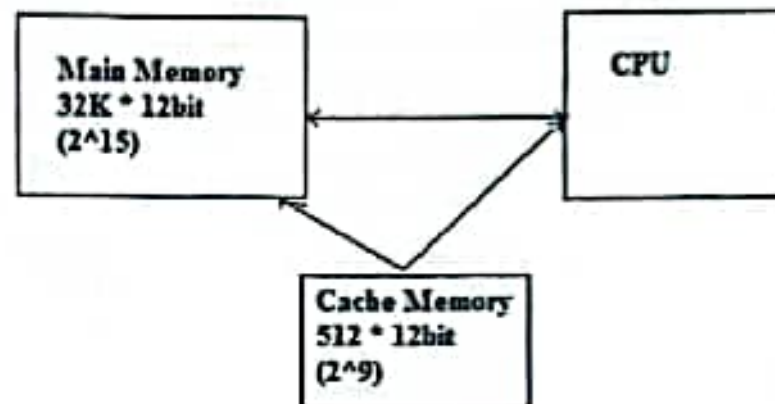
Cache memory

- The basic characteristic of cache memory is its fast access time,
- Therefore, very little or no time must be wasted when searching the words in the cache
- The transformation of data from main memory to cache memory is referred to as a **mapping** process, there are three types of mapping:
 - Associative mapping
 - Direct mapping
 - Set-associative mapping



Cache memory

- To help understand the mapping procedure, we have the following example:



Associative mapping

- The fastest and most flexible cache organization uses an associative memory
- The associative memory stores both the address and data of the memory word
- This permits any location in cache to store any word from main memory
- The address value of 15 bits is shown as a five-digit **octal** number and its corresponding 12-bit word is shown as a four-digit octal number

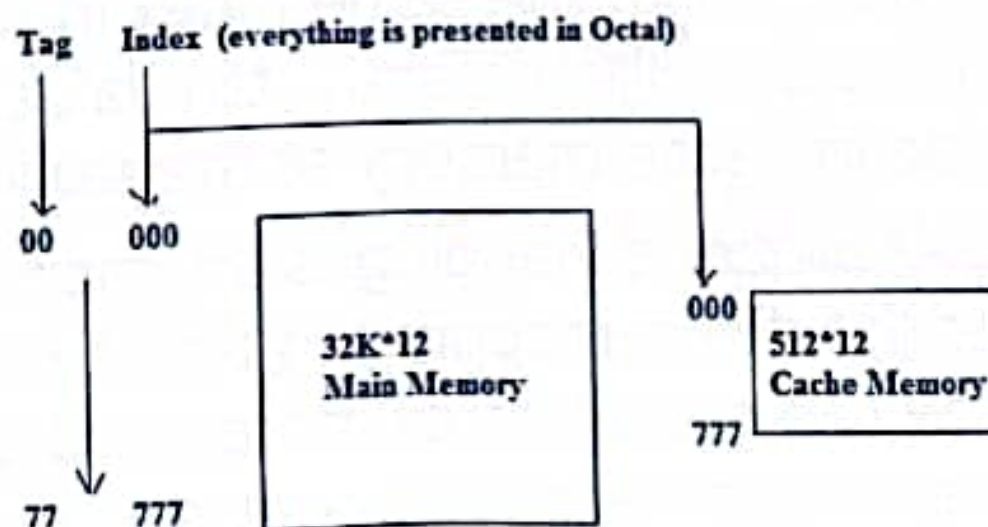


Direct Mapping

- Associative memory is expensive compared to RAM
- In general case, there are 2^k words in cache memory and 2^n words in main memory (in our case, $k=9$, $n=15$)
- The n bit memory address is divided into two fields: k -bits for the index and $n-k$ bits for the tag field



Direct Mapping



Direct Mapping

Memory Address	Memory Data	Index Address	Tag	Data
00000	1220	000	00	1220
00777	2340	111	01	2222
01000	3450			
01111	2222			
01777	4560	777	02	6710
02000	5670			
02777	6710			

Set-Associative Mapping

- The disadvantage of direct mapping is that two words with the same index in their address but with different tag values cannot reside in cache memory at the same time
- Set-Associative Mapping is an improvement over the direct-mapping in that each word of cache can store two or more word of memory under the same index address



Set-Associative Mapping

Memory Address Memory Data

00000	1220
00777	2340
01000	3450
01111	2222
01777	4560
02000	5670
02777	6710

Index Address Tag Data Tag Data

000	01	3450	02	5670
111	01	2222		
777	02	6710	00	2340



Set-Associative Mapping

- In the slide, each index address refers to two data words and their associated tags
- Each tag requires six bits and each data word has 12 bits, so the word length is $2 \times (6 + 12) = 36$ bits